



CLOUD COMPUTING



“If your phone is lost, why are your Google Photos still safe?”

“How can you get your contact list?”

Think: Where are the photos, contacts actually stored?



You Already Use Cloud



Examples you use every day:

- Google Drive, Gmail, YouTube, Facebook
- Google Photos, Canva, Dropbox, OneDrive
- Google Docs, Microsoft 365 online, ChatGPT



What is Cloud Computing?



- ☐ Cloud computing means using computing services through the Internet.
- ☐ Storage, software, servers, databases and networking are provided from remote servers.
- ☐ Easy line: using another powerful computer through the Internet.



Why Is It Called “Cloud”?



- ☐ The cloud means remote servers connected through the Internet.
- ☐ We do not see the server directly, but we use its service.
- ☐ Example: Gmail stores your emails on Google’s servers, not only on your phone.



Power of Cloud Computing



Cloud can:

- Store and back up data online
- Run software without installation
- Host websites, apps and databases
- Support AI, teamwork and remote work
- Handle thousands of users at the same time



Main Characteristics



☐ Resource Flexibility and Scalability

Resources can increase or decrease as needed.

☐ On-Demand Services

Service is available whenever users need it.

☐ Pay as You Go

Users pay only for the service they use.



Types of Cloud Computing



Type	Who uses it?	Example
Public Cloud	General public	Google Drive, Gmail, AWS
Private Cloud	One organization	Bank / hospital private system
Hybrid Cloud	Mix of clouds	Private data + public website
*Community Cloud	Specific community	Healthcare / university cloud



Types of Cloud Computing



Cloud Type	The Analogy	Security & Cost	Real-World Example
Public Cloud	 Like a Public Bus.	Shared by everybody, highly affordable, basically provided without any cost.	Google, Amazon.
Private Cloud	 Like a Private Car.	Exclusively yours, controlled and maintained internally, secure.	NASA Nebula cloud.
Hybrid Cloud	 Renting a Microbus for a trip.	Mixes private security with public scale. Allows excessive amount of resources to be shared.	Netflix.
Community Cloud	 A shared Neighborhood Van.	Multiple local groups sharing one system with specific legal rights.	Healthcare community cloud.



Service Models of Cloud



IaaS = Infrastructure as a Service

PaaS = Platform as a Service

SaaS = Software as a Service

NaaS = Network as a Service

CaaS = Cloud User as a Service



IaaS



Infrastructure as a Service

Cloud company provides virtual server, storage, CPU and networking.

Example: Amazon EC2, Azure Virtual Machine.

Easy line: renting computer infrastructure.



PaaS



Platform as a Service

Cloud company provides a ready platform for developers to build and run apps.

Example: Google App Engine, Azure App Service, Heroku.

Easy line: renting an app-building platform.



SaaS



Software as a Service

Users use ready-made software through the Internet.

Examples: Gmail, Google Docs, Canva, Dropbox, Microsoft 365 online.

Most students use SaaS every day.



NaaS



Network as a Service

Cloud provides network services like VPN, bandwidth, virtual network and secure connection.

Example: secure connection between Dhaka and Chattogram office.



CaaS



Cloud User as a Service

Users enter cloud computing through client devices and browsers.

Devices: computer, laptop, tablet, smartphone.

Browsers: Chrome, Firefox, Edge.



Advantages of Cloud Computing



- Lower operating cost
- Access from anywhere
- Automatic software updates
- Easy sharing and teamwork
- Fast backup and recovery
- Scalable for small or large business



Disadvantages of Cloud Computing



- Requires Internet connection
- Service may stop if provider has a problem
- Privacy and data ownership issues
- Risk of hacking or cyber attack
- Less control over data location and processing



Opportunity in Bangladesh



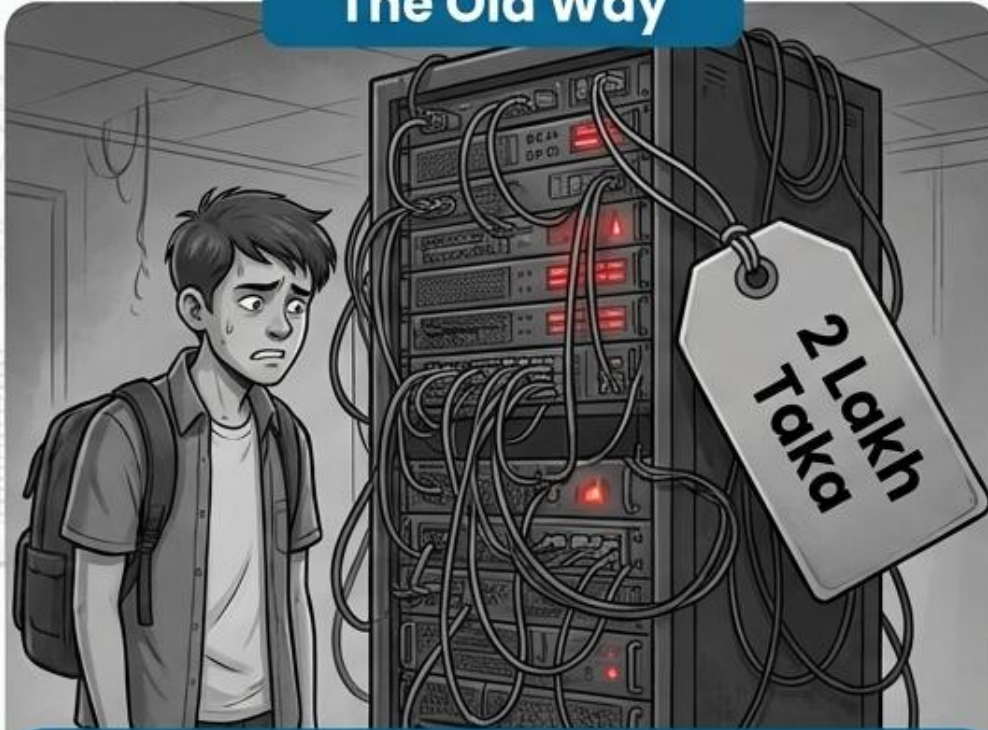
- ☐ Cloud helps entrepreneurs start with low investment.
- ☐ Business ideas: Online shop, coaching center, restaurant order system, digital marketing agency, software startup, freelancing service.
- ☐ Start small, grow when demand increases.



Opportunity in Bangladesh



The Old Way



In the past, starting an IT business meant buying physical servers, managing license keys, and needing immense capital just to host a website.

The Cloud Way



Today, a student can launch an e-commerce site or an educational platform using a \$5/month cloud server.



Can Cadets Build Cloud?



Yes, step by step:

1. Local file sharing in school lab
2. Personal cloud using Nextcloud/ownCloud
3. Host a website on cloud
4. Build online quiz or attendance system
5. Learn Linux, server, database, security



Can Cadets Build Cloud?



Step 1: The Hardware

Take an old, unused computer at home that you thought was garbage.

Nextcloud



Step 2 & 3: Software & Network

Install free, open-source software like Nextcloud onto the PC and connect it to your Wi-Fi router.



The Result

Access, save, and share files from your smartphone straight to your old PC from anywhere on your network.